

Unit 11 – TF-IDF

CS 201 - Data Structures II

Spring 2024

Habib University

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Bag of words

1. The sun is shining
2. The weather is sweet
3. The sun is shining and the weather is sweet

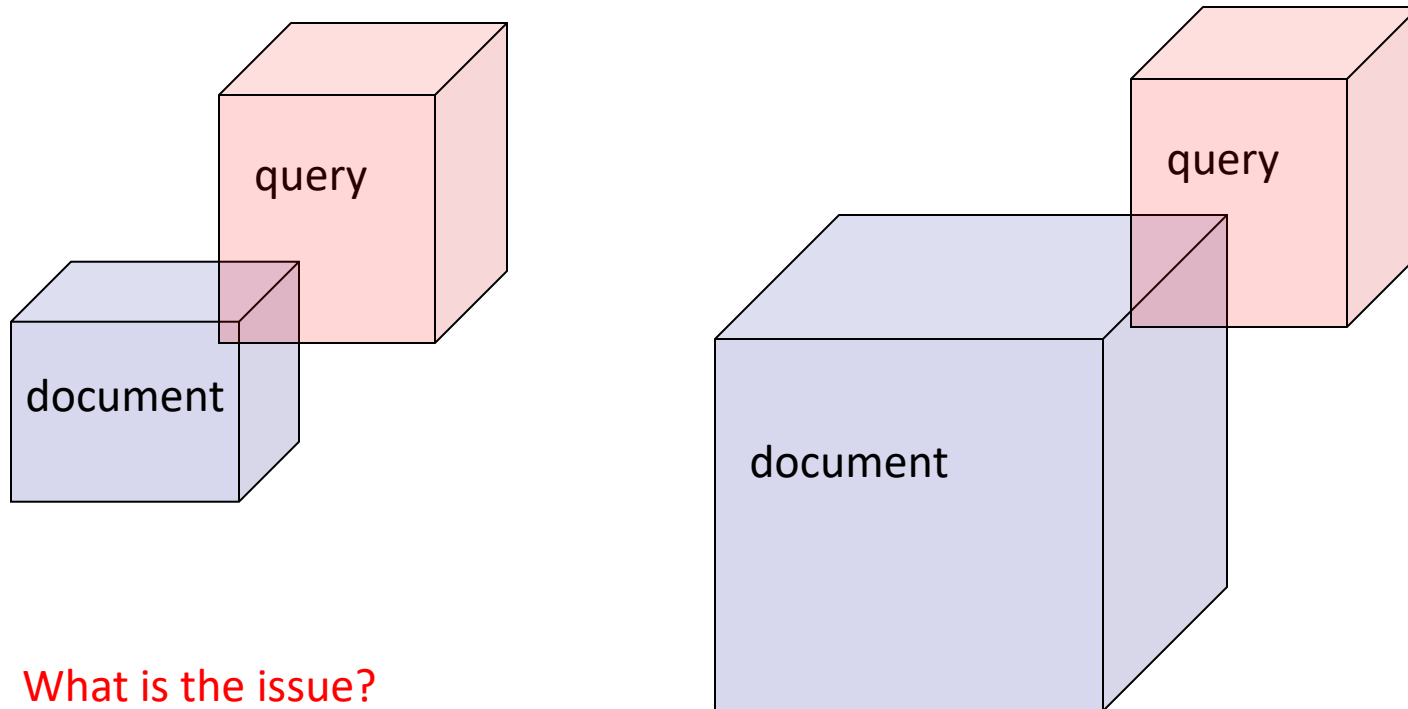
{'the': 5, 'shining': 2, 'weather': 6, 'sun': 3, 'is': 1, 'sweet': 4, 'and': 0}

[[0 1 1 1 0 1 0]

[0 1 0 0 1 1 1]

[1 2 1 1 1 2 1]]

Some things to be careful of...



What is the issue?

Need some notion of the length of a document

Term Frequency

- In document d , the frequency represents the number of instances of a given word t .

$$tf_{i,j} = \frac{n_{i,j}}{\sum_k n_{k,j}}$$

Pitfall with term frequency

← Tweet



The most popular word in each state

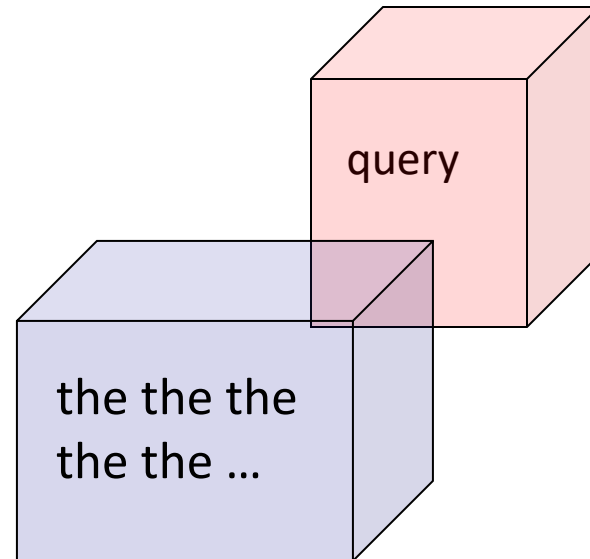
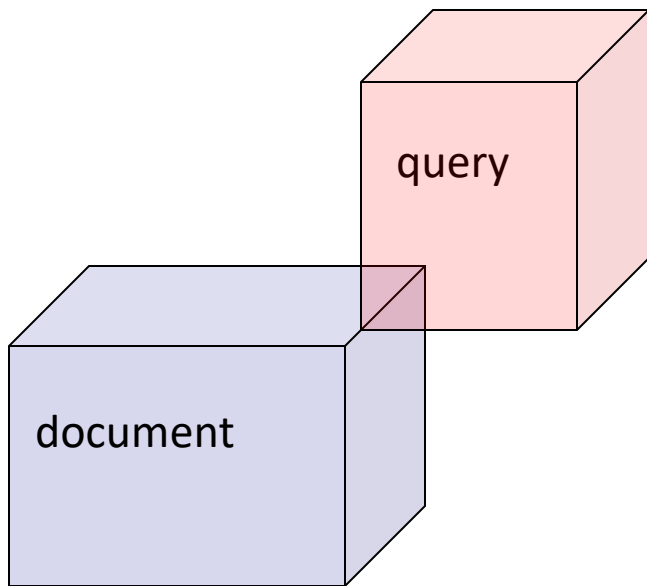


Most frequent word on twitter in each state in the United States .

[Terrible Maps on Twitter: "The most popular word in each state](https://t.co/LY7LewNohn)

[https://t.co/LY7LewNohn" / Twitter](https://t.co/LY7LewNohn)

Some things to be careful of...



Need some notion of the importance of words

Term importance

- Rare terms are more informative than frequent terms
 - Recall stop words
- Consider a term in the query that is rare in the collection
- How to quantify rareness of a term?

Document frequency

- Terms that occur in many documents are weighted less, since overlapping with these terms is very likely
 - In the extreme case, take a word like **the** that occurs in EVERY document
- Terms that occur in only a few documents are weighted more

Inverse Document Frequency (IDF)

- Inverse Document Frequency (IDF)

$$idf_i = \log \frac{|D|}{|d : t_i \in d|}$$

- Calculates how common a word is across documents. Most common terms are less significant.

TF-IDF

- **TF-IDF (term frequency-inverse document frequency)** is a statistical measure that evaluates how relevant a word is to a document in a collection of documents.

$$\text{tf-idf}_{i,j} = \text{tf}_{i,j} \times \text{idf}_i$$

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{'the': 5, 'shining': 2, 'weather': 6, 'sun': 3, 'is': 1, 'sweet': 4, 'and': 0}

```
[ [0 1 1 1 0 1 0]
  [0 1 0 0 1 1 1]
  [1 2 1 1 1 2 1] ]
```

TF-IDF

1. The sun is shining
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```
{'the': 5, 'shining': 2, 'weather': 6, 'sun': 3, 'is': 1, 'sweet': 4, 'and': 0}
```

```
[ [ 0.      0.43  0.56  0.56  0.      0.43  0.    ]  
  [ 0.      0.43  0.      0.      0.56  0.43  0.56]  
  [ 0.4     0.48  0.31  0.31  0.31  0.48  0.31] ]
```

Interpreting TF-IDF

In other words, $\text{tf-idf}_{t,d}$ assigns to t a weight in document d that is:

- highest when t occurs many times within a small number of documents (thus lending high discriminating power to those documents);
- lower when the term occurs fewer times in a document, or occurs in many documents (thus offering a less pronounced relevance signal);
- lowest when the term occurs in virtually all documents.

Exercise

	Doc1	Doc2	Doc3
car	27	4	24
auto	3	33	0
insurance	0	33	29
best	14	0	17

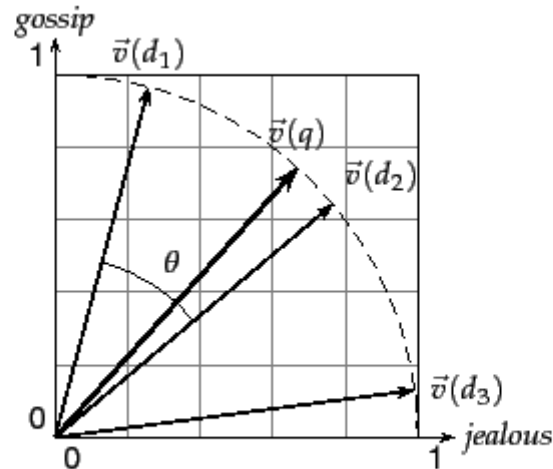
- Compute TF-IDF score of each term in this corpus.

Scoring

The score of a document d is the sum over all query terms that occurs in d .

$$\text{Score}(q, d) = \sum_{t \in q} \text{tf-idf}_{t,d}.$$

Measuring Similarity



► **Figure 6.4** Cosine similarity illustrated. $\text{sim}(d_1, d_2) = \cos \theta$.

Cosine Similarity

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{\sum_{i=1}^n \mathbf{a}_i \mathbf{b}_i}{\sqrt{\sum_{i=1}^n (\mathbf{a}_i)^2} \sqrt{\sum_{i=1}^n (\mathbf{b}_i)^2}}$$

Example

An example is measuring the similarity between documents based on word counts:

Document	Advertising	Auto	Car	Detroit	Engine	Germany	Sales
<i>a</i>	5	88	123	43	35	0	36
<i>b</i>	71	125	42	76	0	27	88

$$a \cdot b = (5 \times 71) + (88 \times 125) + \dots + (36 \times 88) = 22957$$

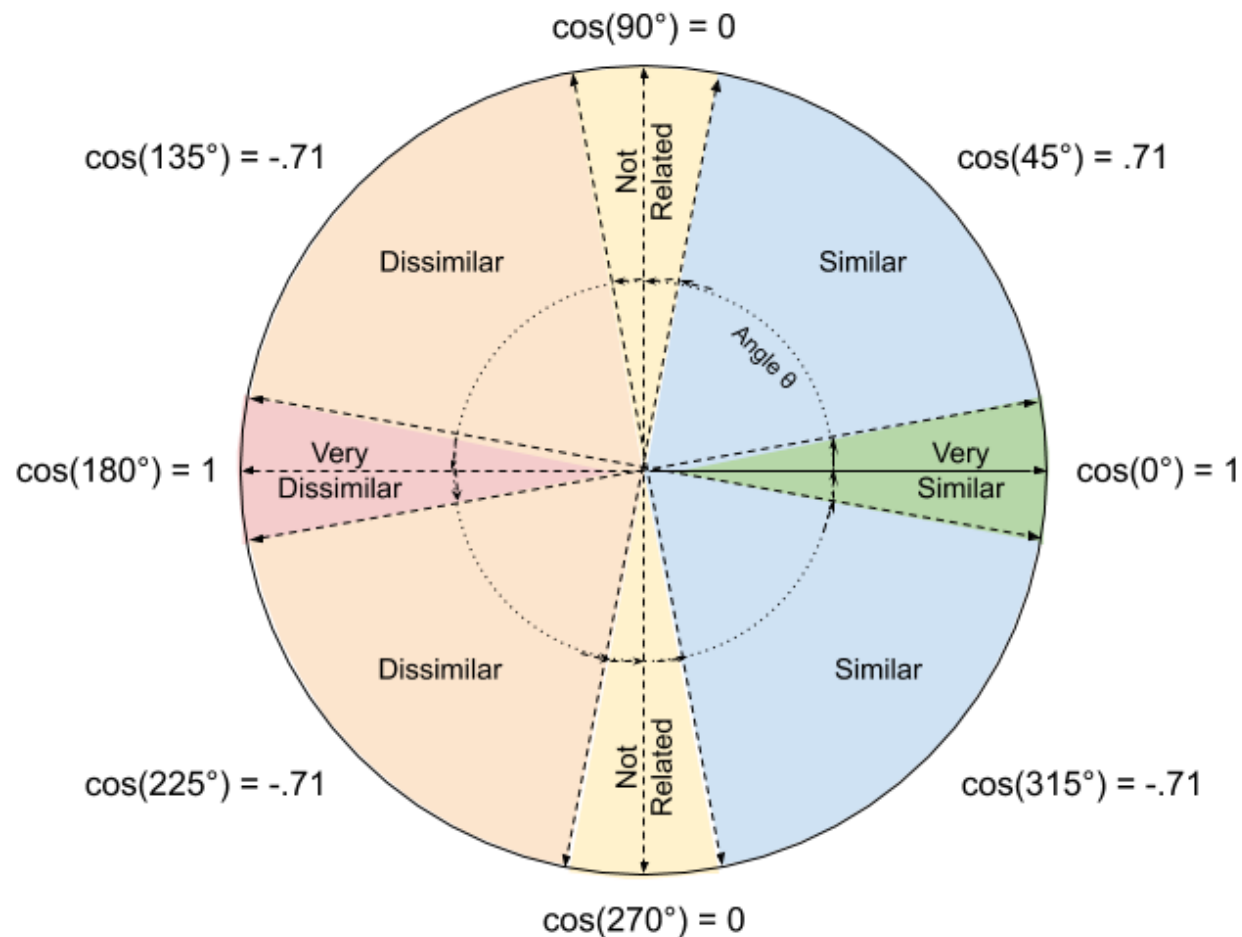
$$\|a\| = \sqrt{5^2 + 88^2 + 123^2 + 43^2 + 35^2 + 0^2 + 36^2} = 165.13$$

$$\|b\| = \sqrt{71^2 + 125^2 + 42^2 + 76^2 + 0^2 + 27^2 + 88^2} = 191.52$$

$$\|a\| \|b\| = 165.13 \times 191.52 = 31626$$

$$\text{cosine similarity} = \cos \theta = \frac{a \cdot b}{\|a\| \|b\|} = \frac{22957}{31626} = .73$$

Interpreting Results



The Three Documents and Similarity Metrics



Considering only the 3 words from the above documents: 'sachin', 'dhoni', 'cricket'

Doc Sachin: Wiki page on Sachin Tendulkar

Dhoni - 10
Cricket - 50
Sachin - 200

Doc Dhoni: Wiki page on Dhoni

Dhoni - 400
Cricket - 100
Sachin - 20

Doc Dhoni_Small: Subsection of wiki on Dhoni

Dhoni - 10
Cricket - 5
Sachin - 1

Document - Term Matrix (Word Counts)

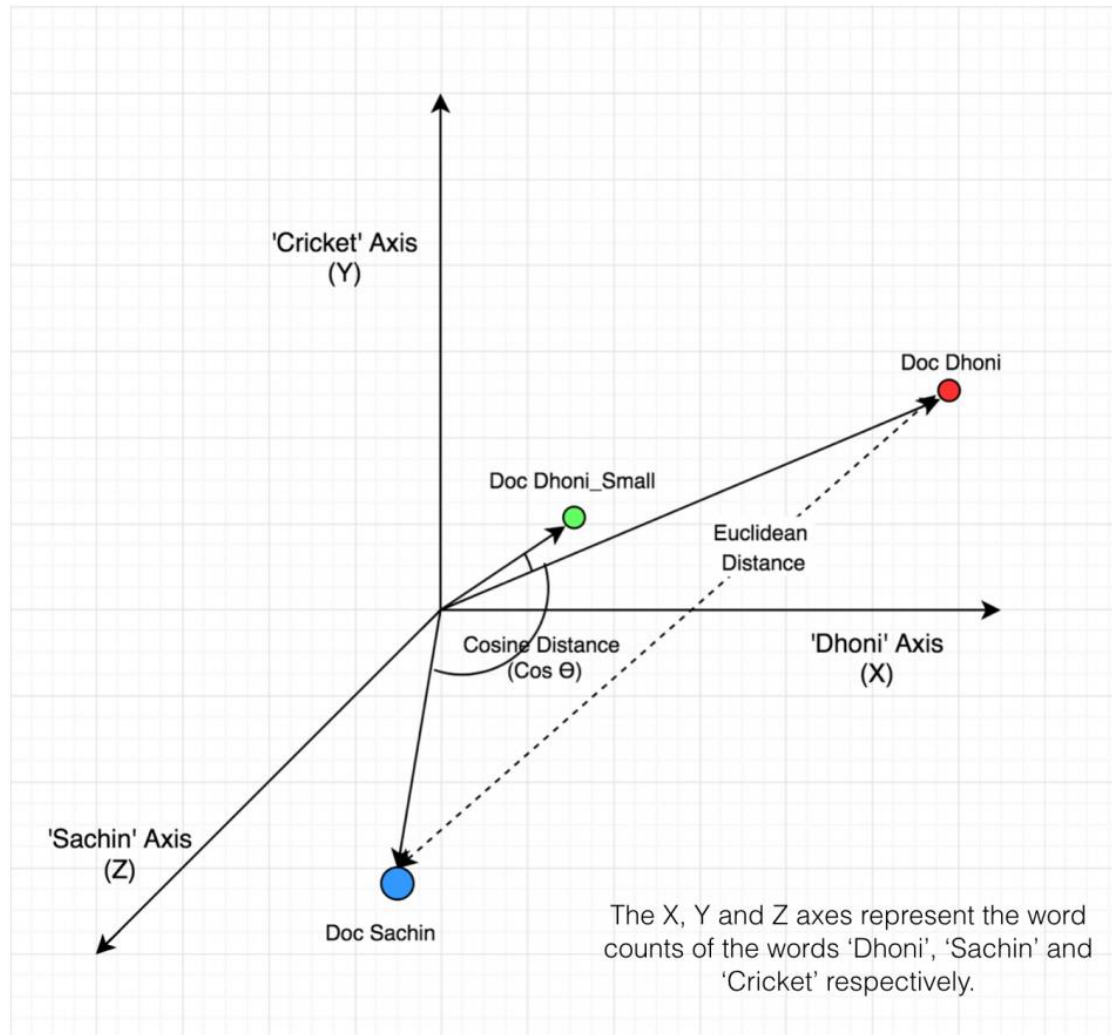
Word Counts	"Dhoni"	"Cricket"	"Sachin"
Doc Sachin	10	50	200
Doc Dhoni	400	100	20
Doc Dhoni_Small	10	5	1



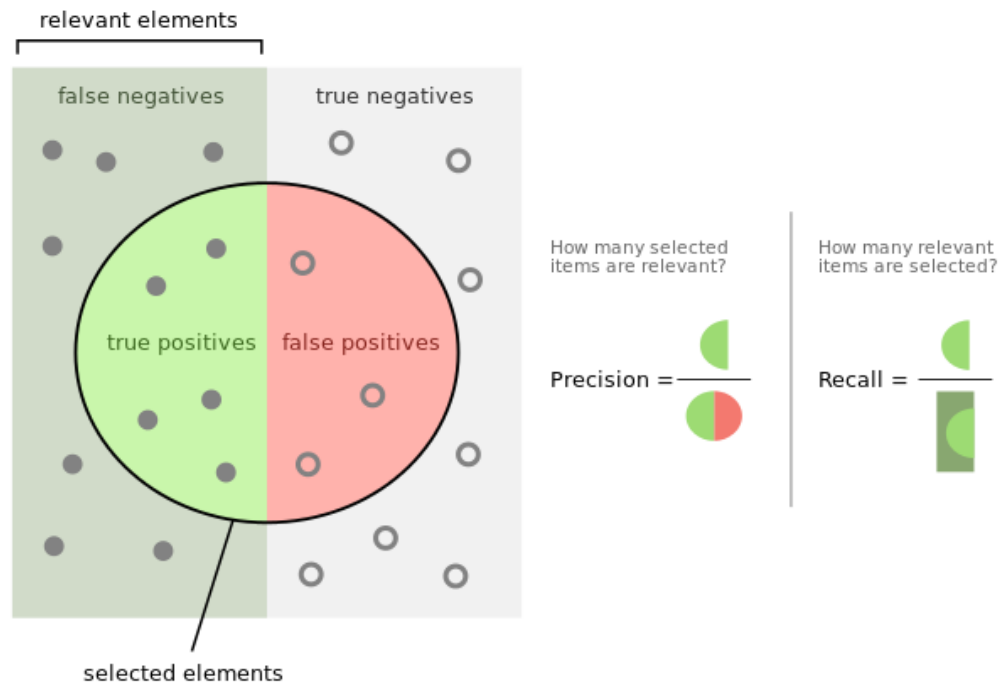
Similarity Metrics

Similarity or Distance Metrics	Euclidean distance	Cosine Similarity
Doc Sachin & Doc Dhoni	432.4	0.15
Doc Dhoni & Doc Dhoni_Small	204.0	0.23
Doc Sachin & Doc Dhoni_Small	401.85	0.77

Projection of Documents in 3D Space



Precision vs Recall



https://en.wikipedia.org/wiki/Precision_and_recall

Precision vs Recall

- It is possible to interpret precision and recall not as ratios but as probabilities.
- Precision is the probability that a (randomly selected) retrieved document is relevant.
- Recall is the probability that a (randomly selected) relevant document is retrieved in a search.

Recall and Precision

Actual	Prediction
T	T
T	F
F	T
F	F
F	T
T	T
T	T
T	F
F	T
T	T

Recall and Precision

Relevant	Retrieved
T	T
T	F
F	T
F	F
F	T
T	T
T	T
T	F
F	T
T	T

- Recall = 4 / 6

Recall and Precision

Relevant	Retrieved
T	T
T	F
F	T
F	F
F	T
T	T
T	T
T	F
F	T
T	T

- Recall = $4 / 6$
- Precision = $4 / 7$
- F-Measure = $8 / 13$

Thanks